

Nonequilibrium landscape of amyloid-beta and calcium ions in application to Alzheimer's disease

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Amyloid-beta ($A\beta$) peptides and calcium ions (Ca^{2+}) serve critical roles in various physiological processes inside neurons, but their dysregulation is involved in the development of neurological disorders, including Alzheimer's disease. Particularly, $A\beta_{42}$ peptides alter Ca^{2+} homeostasis, resulting in synaptic dysfunction and neuronal damage. On the other hand, Ca^{2+} ions can affect $A\beta$ aggregation and clearance routes, either worsening or improving AD pathogenesis. We have considered a deterministic model for $A\beta$ - Ca^{2+} interaction and studied its noise-induced dynamics. The stochastic sensitivity analysis helps to find the confidential ellipses for the stable attractors of the stochastic system. In addition, we have developed a network model based on the interaction between $A\beta$ peptides and Ca^{2+} ions to comprehend their dynamics in the brain connectome better. The probabilistic evolution approach characterizes the dynamics of the stochastic network model and estimates the potentials required to escape from one attractor to another. The numerical simulations suggest that the more connected node requires more energy to switch its state, and the noise intensity plays a crucial role in such cases. Furthermore, this study reveals that external perturbations on $A\beta$ - Ca^{2+} dynamics uncover novel targets for drug development and provide insights into the complex interplay between protein misfolding and synaptic dysfunction in Alzheimer's disease pathology.

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I. INTRODUCTION

Alzheimer's disease (AD) is a neurological illness that causes cognitive decline and memory loss over time. It is the most frequent cause of dementia, a medical condition defined by a major decline in cognitive ability that interferes with daily life. Alzheimer's is named for Dr. Alois Alzheimer, who initially identified the disease in 1906 [1]. It predominantly affects older adults, with the risk increasing with age. However, early-onset Alzheimer's can occur in individuals under sixty-five [2]. AD normally proceeds in phases, starting with mild memory impairment and advancing to severe cognitive decline and loss of independence. Common symptoms include forgetfulness, issues performing routine tasks, confusion, difficulties with language, and changes in mood or behavior [3]. As the condition progresses, people may have difficulty performing daily activities like eating, dressing, and personal hygiene. The prevalence of Alzheimer's is expected to rise due to the global population ages, which poses significant healthcare and socioeconomic challenges. Despite substantial studies, there is presently no cure for Alzheimer's disease; nevertheless, several treatments focus on symptom management and improving the quality of life for individuals and caregivers [4].

The exact causes of Alzheimer's are not fully understood; several key factors have been identified as potential contributors to its development, such as amyloid-beta ($A\beta$) accumulation, tau protein and neurofibrillary tangles, oxidative stress, neuronal calcium dysregulation, etc. In AD, understanding the role of $A\beta$ is critical for developing targeted medicines and interventions that can delay or stop the progression of this devastating neurodegenerative condition [5]. $A\beta$ is a peptide that is derived from the larger amyloid precursor protein (APP). Normally, APP undergoes sequential cleavage by enzymes called beta and gamma secretases, resulting in the generation of $A\beta$ peptides [6]. In AD, however, abnormal processing leads to the accumulation of $A\beta$, particularly $A\beta_{42}$, which has a greater tendency to form aggregates [7]. $A\beta$ peptides have a propensity to aggregate and form insoluble plaques in the brain, called amyloid plaques, consisting of tangled fibrils of $A\beta$. The presence of these plaques is a hallmark neuropathological feature of AD [8–10], and its dynamics have been considered in this study.

Neuronal cells rely on closely regulated calcium concentrations for a variety of physiological processes, including neurotransmitter release, signal transduction, and gene expression [11,12]. Any disruptions of calcium homeostasis can cause neuronal malfunction and cell death [13]. This type of abnormal calcium signaling occurs in neurons for AD [14]. Therefore, understanding the role of calcium ions in AD pathogenesis reveals possible therapeutic targets for the development of new treatment techniques. Modulating

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calcium signaling pathways may provide therapeutic benefits in AD patients by reducing neuronal damage and preserving cognitive function [15,16]. Furthermore, calcium imaging techniques, such as the use of fluorescent dyes, have been used in research to investigate changes in intracellular calcium levels in neurons affected by Alzheimer's [17,18]. These approaches assist researchers in understanding the dynamics of calcium signaling in the setting of the disease.

There is evidence that a peptide linked to AD affects calcium homeostasis in neurons [19]. $A\beta$ can damage cell membrane integrity, leading to an aberrant influx of calcium ions into neurons [20]. This abnormal intracellular calcium level contributes to a variety of downstream effects, including excitotoxicity, synaptic malfunction, and mitochondrial impairment. $A\beta$ in the brain triggers a cascade of processes that raise cytoplasmic calcium, contributing to neurotoxicity and cell death [7]. This work focuses on the complex interactions between $A\beta$ and calcium ions (Ca^{2+}), which helps in developing tailored therapies to restore Ca^{2+} homeostasis and reduce $A\beta$ -induced damage in AD [21–23].

Researchers have been using deterministic and stochastic models for disease progression, and these depend on different assumptions. In the case of a deterministic system, the behavior of the system is entirely determined by its initial conditions and parameter values. Therefore, these models can be used to simulate AD progression based on known biological pathways and other measurable parameters. On the other hand, introducing noise can reveal complex patterns and behaviors resulting from stochastic interactions between system components [24]. These dynamics are especially relevant in systems with nonlinearity, feedback loops, and numerous interacting parts. These types of noise can arise from various sources, such as synaptic variability or external environmental factors [25,26]. The stochastic sensitivity analysis has been applied to see the behavior of the attractors in the presence of noise. Moreover, noise-induced dynamics can facilitate the exploration of the system's robustness, resilience, and adaptability in the face of perturbations.

Understanding how noise influences network dynamics is crucial for deciphering the brain's computational principles. It can gain insights into fundamental principles governing complex biological systems and develop strategies to modulate network behavior for various applications, including neuro-engineering and clinical interventions. In network models of $A\beta$ and Ca^{2+} interactions, incorporating noise can reveal emergent behaviors that may not be apparent in deterministic models. In addition, the network model helps to explore the dynamics of the $A\beta$ and Ca^{2+} in different brain regions. The Fokker-Planck equation helps to find the time evolution of the probability density function of the stochastic system [27]. In AD, it can provide the development of cognitive states over time, taking into account both deterministic (e.g., disease progression pathways) and stochastic elements (e.g., variability in symptom onset and progression among people). Nevertheless, the noise and the degree of each node can affect the resulting steady-state distribution, which can give more insights into the disease state. As a result, the noise-induced network model allows us to determine the influence of noise on the brain connectome, which aids in developing local therapy options in the brain.

II. MATHEMATICAL MODELS

Mathematical models in AD research play a crucial role in understanding the complex dynamics of the disease. These models are intended to depict the underlying biological processes, interactions, and factors contributing to AD's progression. In addition, they assist in predicting the effects of potential interventions and understanding the impact of different factors on disease outcomes.

A. Deterministic system

As a first step, we consider a deterministic model for Alzheimer's disease to analyze the positive feedback loop between amyloid-beta ($A\beta$) and intracellular calcium ions (Ca^{2+}) as [21,23]:

$$\begin{aligned}\frac{du}{dt} &= a_1 + a_2 f(v) - k_1 u, \\ \frac{dv}{dt} &= \frac{1}{\epsilon}(b_1 + b_2 g(u) - k_2 v),\end{aligned}\tag{1}$$

where u and v are the concentrations of amyloid-beta and intracellular calcium ions, respectively. Here, a_1 and k_1 are the basic synthesis and clearance rates of amyloid-beta, respectively. The term $a_2 f(v)$ is the positive feedback of Ca^{2+} acting on $A\beta$ with a_2 as the positive strength and we choose $f(v) = v^2/(v^2 + \alpha^2)$ as the Holling type-III function. For the Ca^{2+} equation, the terms b_1 and k_2 represent the flow rate into the cytoplasm and the natural decay rate, respectively. The term $b_2 g(u)$ represents the positive feedback of $A\beta$ acting on Ca^{2+} with the strength b_2 . We choose $g(u) = u$ because the concentration of Ca^{2+} changes quickly in the cell. Furthermore, the parameter ϵ ($0 < \epsilon \ll 1$) is considered to incorporate the different time scales of the onset of amyloid beta (in years) and calcium ions (in seconds).

B. Noise-induced system

The consideration of random effect in studying biological models has great significance, as random uncertainty is an inevitable factor in developing organic tissues and biological systems. In addition, studying random uncertainty in Alzheimer's disease is crucial because it manifests differently in each individual. This further accounts for different variations in disease progression and understanding the disease status more appropriately, which can help improve the effectiveness of therapeutic interventions. Furthermore, these noises can be the intracellular and extracellular environment subject to various perturbations that affect the concentration balance of amyloid-beta and calcium ions in the brain. Before going to the network model in the brain connectome, we first include the stochastic term in the temporal model (1) as follows:

$$\begin{aligned}du &= \left(a_1 + \frac{a_2 v^2}{v^2 + \alpha^2} - k_1 u\right)dt + \xi_1 dB_1, \\ dv &= \frac{1}{\epsilon}(b_1 + b_2 u - k_2 v)dt + \frac{\xi_2}{\sqrt{\epsilon}}dB_2,\end{aligned}\tag{2}$$

where B_1 and B_2 are the Gaussian white noises with intensities ξ_1 and ξ_2 , respectively. For simplicity, we choose

$\xi_1 = \xi_2 = \xi$. Due to the random disturbances, the stable equilibrium point faces the concentrated oscillations around it, and in this case, the stochastic states form a stationary probabilistic distribution $\rho(\mathbf{y}, \xi)$ where $\mathbf{y} = (u, v)$ [28]. This distribution can be found by studying the corresponding Fokker-Planck-Kolmogorov equation [29]. Based on the theory of quasipotentials, this probability distribution can be approximated by the Gaussian distribution for low noise intensity [30],

$$\rho(\mathbf{y}, \xi) \approx K \cdot \exp\left(-\frac{\Psi(\mathbf{y})}{\xi^2}\right),$$

where the quasipotential can be expressed as $\Psi(\mathbf{y}) = -\lim_{\xi \rightarrow 0} \xi^2 \log \rho(\mathbf{y}, \xi)$ [31]. In addition, for an exponentially stable equilibrium point $\mathbf{y}_s = (u_s, v_s)$, the quadratic approximation of the potential $\Psi(\mathbf{y}) \approx \frac{1}{2}(\mathbf{y} - \mathbf{y}_s, \mathbf{W}^{-1}(\mathbf{y} - \mathbf{y}_s))$ can be used to find the stationary distribution $\rho(\mathbf{y}, \xi)$, where $\mathbf{W}$ is a positive definite second-order matrix satisfies the matrix equation

$$\mathbf{G}\mathbf{W} + \mathbf{W}\mathbf{G}^\top + \mathbf{S} = \mathbf{0},$$

with $\mathbf{G}$ is the Jacobian matrix at the equilibrium $\mathbf{y}_s$ and $\mathbf{S} = \text{diag}\{1, 1/\epsilon\}$. Here, $\mathbf{W}$ is called the stochastic sensitivity matrix for the equilibrium $\mathbf{y}_s$, which helps to find the stationary distributed random states of the stochastic system around the deterministic equilibrium $\mathbf{y}_s$. In the two-dimensional case, the confidence ellipse $(\mathbf{y} - \mathbf{y}_s, \mathbf{W}^{-1}(\mathbf{y} - \mathbf{y}_s)) = 2k^2\xi^2$ determines the random states around the equilibrium $\mathbf{y}_s$ where $k = -\ln(1 - \mathcal{P})$ with $\mathcal{P}$ as the fiducial probability (probability of random states belong to this ellipse).

C. Spatial extension in a brain network

There are several advantages to exploring the dynamics of a network. For instance, when disease or influence propagates through a network, researchers can predict systemic changes and create plans to reduce unfavorable effects and increase favorable ones. Furthermore, it allows us to identify some key components and understand their influence on the overall system. In general, amyloid beta and calcium ions diffuse in the brain region. Therefore, the dynamics of the amyloid-beta and the calcium ion concentrations can be captured by extending the model (1) into the network model in the brain connectome as [32,33]:

$$\begin{aligned} \frac{du_j}{dt} &= -\rho_u \sum_{k=1}^N L_{jk} u_j + a_1 + \frac{a_2 v_j^2}{v_j^2 + \alpha^2} - k_1 u_j \\ &\equiv f_j(u_1, \dots, u_N, v_1, \dots, v_N), \\ \frac{dv_j}{dt} &= \frac{1}{\epsilon} \left(-\rho_v \sum_{k=1}^N L_{jk} v_j + b_1 + b_2 u_j - k_2 v_j \right) \\ &\equiv g_j(u_1, \dots, u_N, v_1, \dots, v_N), \end{aligned} \quad (3)$$

for $j = 1, 2, 3, \dots, N$. Here, N is the number of nodes in the brain connectome, L_{jk} are the entries of the Laplacian matrix $\mathbf{L}$ for a given network, and ρ_u and ρ_v are the diffusion coefficients of amyloid beta and calcium ions, respectively [32,33]. Now, we include the stochastic dynamics in the network

model (3) as

$$\begin{aligned} du_j &= f_j(u_1, \dots, u_N, v_1, \dots, v_N)dt + \eta_{u_j} dB_{u_j}, \\ dv_j &= g_j(u_1, \dots, u_N, v_1, \dots, v_N)dt + \frac{\eta_{v_j}}{\sqrt{\epsilon}} dB_{v_j}, \end{aligned}$$

for $j = 1, 2, 3, \dots, N$. This system of equations can be written in the form [34]:

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}(t)) + \eta(t), \quad (4)$$

where

$$\begin{aligned} \mathbf{x} &= (u_1, \dots, u_N, v_1, \dots, v_N)^\top, \\ \mathbf{F} &= (f_1, \dots, f_N, g_1, \dots, g_N)^\top, \end{aligned}$$

and η is a $2N$ -dimensional vector representing independent Gaussian white noise, i.e.,

$$\langle \eta(t) \rangle = 0,$$

$$\langle \eta(t)\eta(s) \rangle = 2d \cdot \mathbf{I}_\epsilon \cdot \delta(t - s),$$

where d is the diffusion coefficient, and $\delta(t)$ is the Dirac delta function. Here, the parameter d measures the level of noise strength, and hence, increasing the diffusion parameter d causes an increase in the noise intensity and vice versa. The matrix $\mathbf{I}_\epsilon$ is a diagonal matrix of order $2N \times 2N$ whose first N -diagonal elements are 1, and the remaining diagonal elements are $1/\epsilon$.

Now, the time evolution of probability distribution for the stochastic system (4) can be described by the Fokker-Planck (FP) equation, and it is given by

$$\begin{aligned} \frac{\partial P(\mathbf{x}, t)}{\partial t} &= - \sum_k \frac{\partial}{\partial x_k} [F_k(\mathbf{x}(t))P(\mathbf{x}, t)] \\ &\quad + d \sum_k \mathbf{I}_\epsilon(k, k) \frac{\partial^2 P(\mathbf{x}, t)}{\partial x_k^2}, \end{aligned} \quad (5)$$

where $P(\mathbf{x}, t)$ is the probability density function of the system state at time t and $\mathbf{I}_\epsilon(k, k)$ is the (k, k) -th element of the matrix $\mathbf{I}_\epsilon$. Here, we assume the diffusion coefficient tensor is homogeneous and isotropic. The system (5) can be written as

$$\frac{\partial P}{\partial t} + \nabla \cdot \mathbf{J}(\mathbf{x}, t) = 0, \quad (6)$$

where $\mathbf{J}(\mathbf{x}, t) = \mathbf{F}P - d \cdot \mathbf{I}_\epsilon \cdot \frac{\partial}{\partial \mathbf{x}} P$ is the probability flux vector measuring the speed of the flow in concentration space $\mathbf{x}$. Furthermore, Eq. (6) represents the conservation law of probability [35]. However, simplifying the FP equation into form (6) for the stochastic model with multiplicative noise terms is challenging, so we use other approaches to solve it; for example, see [36,37]. In many networks, the steady state exists and in that case, $\frac{\partial P}{\partial t} = 0$, which implies from the conservation law of probability (6) that $\nabla \cdot \mathbf{J}(\mathbf{x}, t) = 0$. It is also known that the divergence of $\mathbf{J}$ vanishes at the steady state. Now, the condition $\nabla \cdot \mathbf{J}(\mathbf{x}, t) = 0$ is true for two cases: $\mathbf{J}_{ss} = 0$ and $\mathbf{J}_{ss} \neq 0$.

For the case of $\mathbf{J}_{ss} = 0$ implies $\mathbf{F}P_{ss} = d \cdot \mathbf{I}_\epsilon \cdot \frac{\partial}{\partial \mathbf{x}} P_{ss} \Rightarrow \mathbf{F} = d \cdot \mathbf{I}_\epsilon \cdot \frac{\partial}{\partial \mathbf{x}} P_{ss}/P_{ss} = -d \cdot \mathbf{I}_\epsilon \cdot \frac{\partial}{\partial \mathbf{x}} (-\ln P_{ss}) = d \cdot \mathbf{I}_\epsilon \cdot \frac{\partial}{\partial \mathbf{x}} U$, where $U = -\ln P_{ss}$. Therefore, the driving force $\mathbf{F}$ can be expressed as a gradient of a potential U , which has a link with

a steady-state probability distribution. This further shows that the zero flux reflects the detailed balance condition under which the system is in equilibrium. On the other hand, $\mathbf{J}_{ss} \neq 0$ implies $\mathbf{J}_{ss} = \mathbf{F}P_{ss} - d \cdot \mathbf{I}_\epsilon \cdot \frac{\partial}{\partial \mathbf{x}} P_{ss} \Rightarrow \mathbf{F} = d \cdot \mathbf{I}_\epsilon \cdot \frac{\partial}{\partial \mathbf{x}} P_{ss}/P_{ss} + \mathbf{J}_{ss}/P_{ss} = d \cdot \mathbf{I}_\epsilon \cdot \frac{\partial}{\partial \mathbf{x}} (-\ln P_{ss}) + \mathbf{J}_{ss}/P_{ss} = d \cdot \mathbf{I}_\epsilon \cdot \frac{\partial}{\partial \mathbf{x}} U + \mathbf{J}_{ss}/P_{ss}$. In this case, the driving force decomposed into a gradient of a generalized potential U and steady-state divergent-free curl flux field $\mathbf{J}_{ss}$. Therefore, the nonzero flux reflects the lack of detailed balance in nonequilibrium systems.

The system (5) has $2N$ number of independent variables, and finding the exact solution to this Fokker-Plank equation is not feasible. It is because every variable can have M (say) values, and the dimensionality of the system becomes M^{2N} . We apply the self-consistent mean field approach to reduce the dimensionality problem, and in this case, the probability splits into the product of individual probabilities. This process reduces the dimensionality from exponentially large M^{2N} to polynomially small $M \times 2N$, which is computationally tractable. In addition, it is still challenging to solve the diffusion probability directly, and we simplify the computational task by assuming specific probability distributions based on physical arguments. Here, we assume that the time evolution of the probability density function $p(\mathbf{x}, t)$ satisfies the Gaussian distribution, and it requires two moments, mean $\bar{\mathbf{x}}(t)$ and variance $\Sigma(t)$. For small diffusion coefficient d , the moment equations can be approximated as

$$\begin{aligned} \dot{\bar{\mathbf{x}}}(t) &= \mathbf{F}(\bar{\mathbf{x}}(t)), \\ \dot{\Sigma}(t) &= \Sigma(t)\mathbf{A}^\top(t) + \mathbf{A}(t)\Sigma(t) + 2d \cdot \mathbf{I}_\epsilon, \end{aligned} \quad (7)$$

where $\mathbf{A}(t)$ is the Jacobian matrix of $\mathbf{F}(\mathbf{x})$, and $\mathbf{A}^\top(t)$ is its transpose matrix. In this work, we focus on the behaviors of the system at the steady state. Furthermore, the probability density distribution of the system at the steady state can be expressed as

$$P_{ss}(\mathbf{x}) = \frac{1}{(2\pi)^N |\sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \mu)^\top \sigma^{-1} (\mathbf{x} - \mu) \right\},$$

where μ and σ are the solutions of the system (7) for $t \rightarrow \infty$. If the system has M number of stable attractors, then we use the weighted sum of Gaussian distributions to represent the probability density function of the system as

$$P_{ss}(\mathbf{x}) = \sum_{j=1}^M \phi_j P_{ss}^j(\mathbf{x}),$$

where $P_{ss}^j(\mathbf{x})$ is the probability density function for the j -th stable state and ϕ_j is the corresponding weight. In order to calculate the weights, we first solve the ODE with a large number of random initial conditions, and then the frequency gives the weight of each stable state.

III. RESULTS AND DISCUSSION

Before going to the nonequilibrium dynamics for the network model, we first discuss the dynamics of the deterministic and stochastic models. In this study, we fix all the parameter values used in the system (1) and they are given in Table I.

TABLE I. Synthetic parameter values used for the numerical simulations [23].

Parameter	a_1	a_2	α	k_1	b_1	b_2	k_2	ϵ
Value	0.25	2.89	1.0	0.35	0.11	1.0	5.0	0.01

A. Deterministic dynamics

The deterministic system has three equilibrium points for the fixed parameter set, given in Table I: two are stable, and one is a saddle point. Generally, every stable equilibrium point has a basin of attraction, so the considered system has two basins of attraction for this parametric set. The lower stable attractor is the healthy state, and the higher stable attractor is the pathological state. With an increase in the two rate constants a_2 (γ -secretase) and b_1 (the rate of Ca^{2+} entry), there will be a total disappearance of the low steady-state branch, i.e., the healthy state [21]. Figure 1 depicts such dynamics for the bifurcation parameter a_2 with $b_1 = 0.11$. The deterministic system (1) has a healthy stable state for $a_2 < 2.9844$ and disappears through the saddle-node bifurcation at $a_2^{SN_2} = 2.984$. However, the pathological state appears through another saddle-node bifurcation at $a_2^{SN_1} = 2.8444$ and stays stable for higher values of a_2 . Therefore, the system shows a bistable behavior for the parametric range $a_2^{SN_1} < a_2 < a_2^{SN_2}$ [see Fig. 1]. To see the dynamics of the deterministic system, we have considered three parametric values of a_2 , and their phase portraits are plotted in Fig. 1. For the first case, the system has only the healthy stable state (blue diamond) for $a_2 = 2.5$, and the solution trajectories go to the stable state for different initial conditions [see Fig. 1(a)]. For the second case, the system shows the bistability for $a_2 = 2.89$ where the basin of attractions of the two stable attractors (magenta bullet points) are separated by the separatrix that comes from the unstable manifolds (black-dashed) of the saddle point (red bullet point) [see Fig. 1(b)]. For the final case, the system has only the pathological stable state (black hexagon) for $a_2 = 3$, where the solution trajectories converge to that state for different initial conditions [see Fig. 1(c)]. The same scenario can be observed for the parameter b_1 for $a_2 = 2.89$. Consequently, the presenilin protein mutations associated with familial AD cause a decline in the region of bistability, which reduces the likelihood of a single, stable pathogenic state.

B. Noise-induced temporal dynamics

In the presence of stochastic noise, the stable attractors for the deterministic system may or may not remain stable. Sometimes, the solution for the stochastic system oscillates around the stable attractors, and the area of the oscillating region depends on the noise intensity level. We use the stochastic sensitivity analysis given in Sec. II B to find such regions for the stochastic system (2) around the stable attractors of the deterministic system, and these regions are called confidence ellipses. Figure 2 depicts the confidence ellipses for both the stable attractors for different noise intensities with the fiducial probability $\mathcal{P} = 0.999$. It shows that the area of the confidence ellipses increases with the noise intensity. For low noise intensities, the stochastic states lie inside the

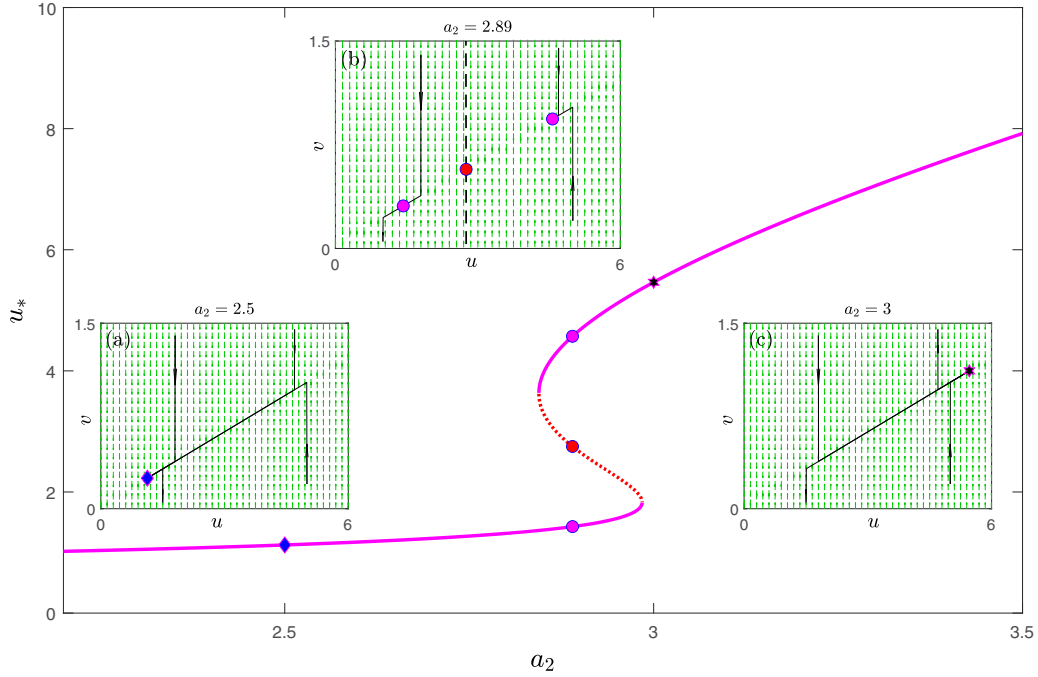


FIG. 1. Bifurcation and phase portraits for the deterministic system (1) for the parameter a_2 : (a) one equilibrium point for $a_2 = 2.5$, (b) three equilibria for $a_2 = 2.89$, and (c) one equilibrium point for $a_2 = 3$.

corresponding ellipses [see Figs. 2(a) and 2(b)], but they can move to the other attractor or switch between attractors for a higher noise intensity [see Fig. 2(c)]. This shows that the nonequilibrium dynamics occur in the system in the presence of higher noise intensity. In general, the trajectories for the deterministic system do not cross the separatrix, but in the presence of stochastic noise, they can cross the separatrix that exists for the deterministic system. This shows that a higher noise level can change one state into another in the system, but maintaining the latter state is very important in such cases as it can switch back to the previous state where it was moved or

into some other state. However, a minimum amount of noise intensity is required to switch the system's state. Therefore, a noise-induced phase transition could be possible, transferring the pathological state into a healthy state.

The densities of the amyloid-beta and calcium ions do not remain homogeneous throughout the brain. Therefore, studying their density distributions in the brain is a critical aspect of understanding neurodegenerative disease progression, monitoring, and developing effective treatments. Theoretically, we have seen that the system's state can be switched to another state with the help of stochastic noise. Now, we focus on this

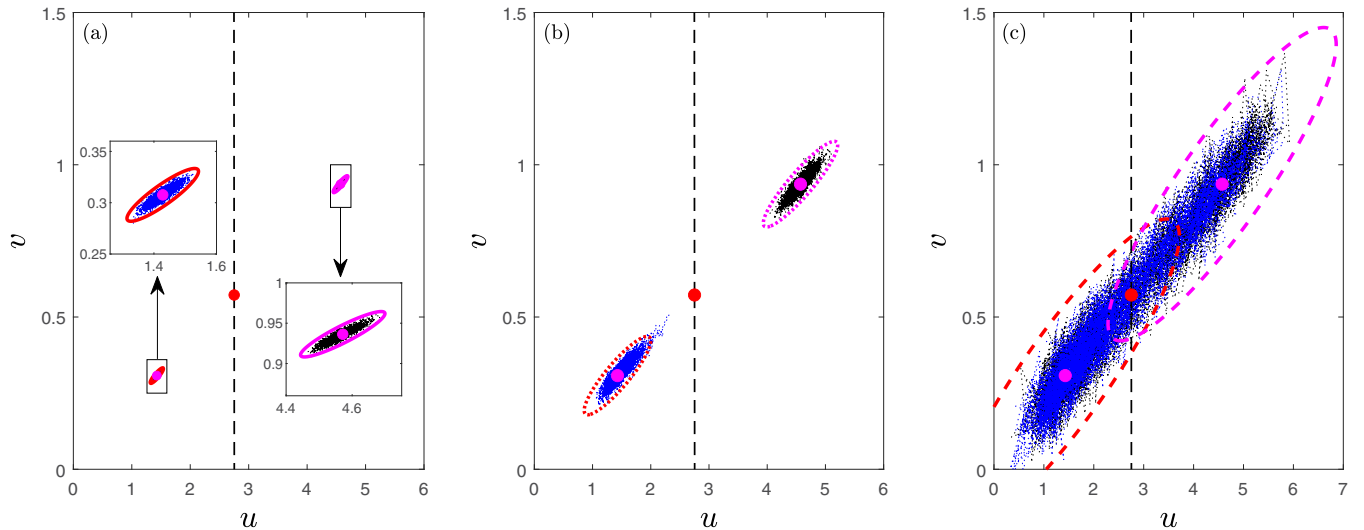


FIG. 2. Confidence ellipses for the stochastic system (2) for $a_2 = 2.89$ with fiducial probability $\mathcal{P} = 0.999$. Solid, dotted and dashed confidential ellipses are for $\xi = 0.01, 0.05$, and 0.2 , respectively. Equilibrium points for the deterministic system (1) are plotted by bullet points, and the separatrix for the saddle point is plotted by the black dashed curve.

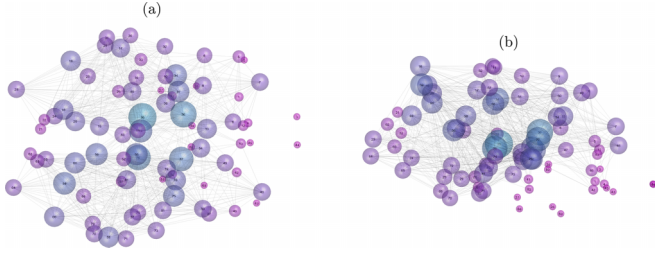


FIG. 3. Three-dimensional views of the integrated brain connectome data with nodes are in the different regions: (a) the axial view and (b) the sagittal view. The size of the nodes is based on their degree.

switching behavior locally in different brain regions with the help of brain connectome data and the potential required to make it happen.

C. Noise-induced dynamics in a brain network

This study includes large-scale brain connectome data consisting of 83 nodes and 790 edges, which is freely available at [38]. These data have been collected by high-angular resolution diffusion imaging method, and each node represents a small area of about 1–1.5 cm² of the gray matter in the anatomically identified regions [39]. Furthermore, there is an edge between two nodes when neural fiber tracts are observed between them. Based on the imported data, the axial and sagittal view of the brain network is presented in Fig. 3. In the integrated data, nodes are connected through edges, and the number of edges is not the same for all the nodes. The total number of edges associated with each node in the network is called the degree of that node. In Fig. 3, the nodes are plotted in different sizes to distinguish their degrees. The smallest and largest sizes in the plot indicate degree zero (isolated node) and the highest degree (the most connected node), respectively, and the other sizes represent the intermediate degrees. Now, we see how the landscape of amyloid-beta and calcium ions changes at different nodes based on their connectivity in the presence of different levels of noise strength.

In the early stage of AD, the brain regions are almost properly connected, but as the disease spreads into the whole brain, the regions become isolated, and communication through the neurons is almost impossible. Hence, we can assume that a less connected brain is more towards the disease state because regions can not communicate themselves frequently due to more breakage of neurons in the brain network. Therefore, the computational study on the dependence of the degree of each node in the brain connectome is crucial in studying the disease. Nevertheless, the considered network model can capture the interaction between the amyloid beta and calcium ions among different nodes due to the presence of the Laplacian matrix $\mathbf{L}$. This Laplacian matrix is based on the mean fibre numbers and their lengths and nevertheless depends on the degree of each node. We find it for the integrated brain connectome data (the procedure is mentioned in [33]). In addition, if a node is disconnected (degree zero), then the interaction of $A\beta$ or Ca^{2+} through the diffusion term (here, it is Laplacian) is not possible and hence only interacts with themselves present at the same node in the brain connectome.

We use the general method described in the earlier section and find the steady-state probability distributions with a self-consistent mean-field approach. The potential landscape of the organisms is defined as $U(\mathbf{x}) = -\ln(P_{ss}(\mathbf{x}))$, where P_{ss} represents the probabilistic distribution at steady state. Due to the multi-dimensionality problem of the state variable $\mathbf{x}$, we present two state variables in plots from 166 variables to map out the landscape by integrating the remaining variables. These two state variables correspond to the amyloid-beta and calcium ions at the same node in the graph. Furthermore, we have used over 10 000 random initial conditions from uniform distributions to prevent local trapping in the solution. We have utilized the Runge-Kutta fourth-order method with a time step of $dt = 0.01$ and 10^6 iterations. This approach helps to find the weights (ϕ_j) of the stable states as the simulations show two individual stable states present in the system for the given parameter values in Table I. The system can have a single stable state for the other parameter values of a_2 [see Fig. 1]. We do not discuss such cases, as this article focuses on how the potential landscape changes in the presence of multiple attractors with Gaussian white noise.

We present the potential landscape at different nodes in the presence of different noise intensities. The upper panel of Fig. 4 depicts the potential landscapes $U(\mathbf{x})$ of the stationary states at four different nodes for the noise intensity $d = 0.001$. These nodes are chosen due to their different degrees. This work considers the dynamics of $A\beta$ and Ca^{2+} at the nodes which have a degree greater than zero because we are interested in the collective change in the brain connectome. Moreover, the potential landscape for the nodes with degree zero reflects the same dynamics as the temporal model because there is no contribution from the Laplacian diffusion terms.

The upper panel of Fig. 4 shows the potential landscapes for the nodes 39, 4, and 36 for $d = 0.001$. These contour plots show that the valleys' shapes and sizes around the potential landscape's stable attractors vary with their degrees, as the degrees of the nodes 39, 4, and 36 are 1, 4, and 42, respectively. In addition, these results show that the nodes with a degree greater than zero have disjoint basins of attractions for the low noise intensity. Furthermore, the more connected node has the most isolated basin of attraction around the stable state in the presence of low noise intensity. We have shown the surface plot of the potential landscapes for $d = 0.1$ for nodes 39, 4 and 36 in Figs. 4(d)–4(f), respectively. In this case, a significant difference in the steady-state probability density function P_{ss} has been observed compared to $d = 0.001$ for each node in the brain connectome. Here, we have selected these nodes by their degrees in increasing order and left the other nodes. However, we will analyze these landscapes for all positive-degree nodes later in this section by incorporating barrier height.

As we see in Fig. 4, the shapes and sizes of the valleys around the stable attractors in the potential landscape are different at different nodes. Any small perturbation in the system around these stable attractors causes oscillation in the $A\beta$ and Ca^{2+} , and they may or may not stay in the range of corresponding basins of attraction in such situations. If the perturbation is sufficiently large, then the solutions can leave from one basin of attraction to another. This escaping behavior depends on the barrier height, which is the potential difference

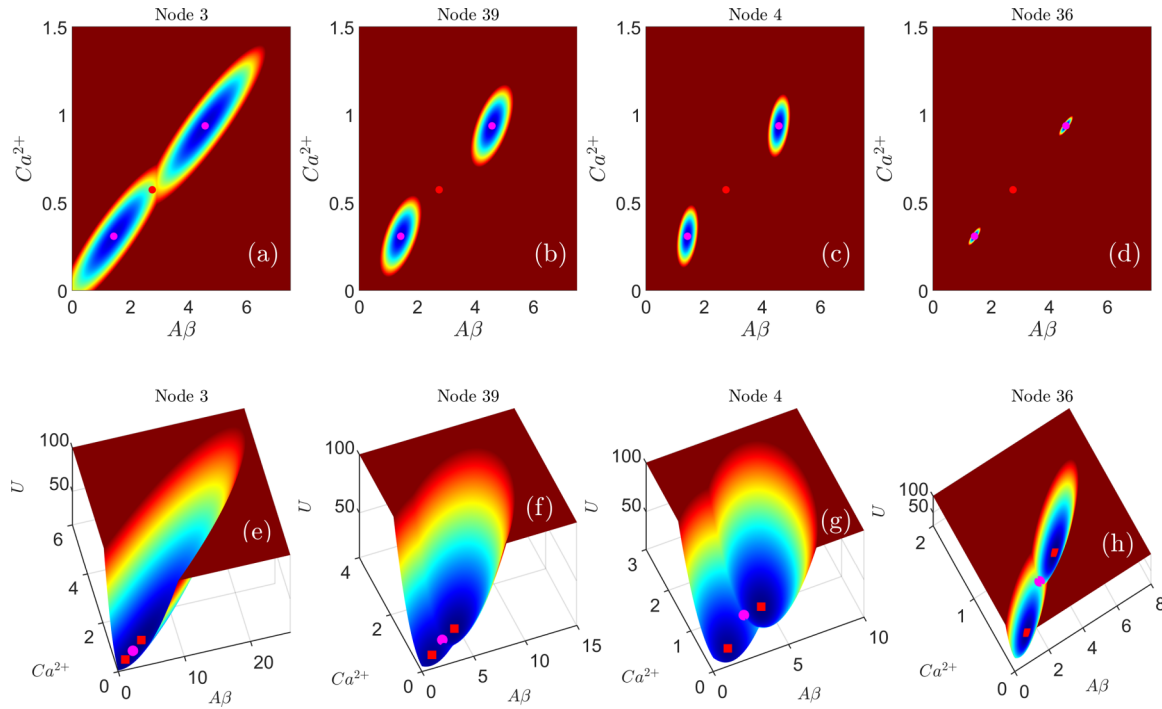


FIG. 4. The contour plot and surface plot of the potential landscape of the stationary state at different nodes in the brain connectome for $d = 0.001$ (upper panel) and $d = 0.1$ (lower panel).

between the potentials at the stable and saddle points. Figure 5 presents a schematic diagram to find this barrier height when two stable attractors are present in a system. In the diagram, we consider two square blue points S_1 and S_2 as the stable states, and the red bullet point U_1 as the saddle point. In addition, the heights H_1 and H_2 are the barrier heights for the stable attractors S_1 and S_2 , respectively. Furthermore, the difference $\Delta U = H_1 - H_2$ is the potential barrier for calculating the transition state or Kramers's escaping rate.

It has shown that the noise intensity affects the potential landscapes of the attractors [see Fig. 4]. In addition, the potential landscape and the barrier height for each node in

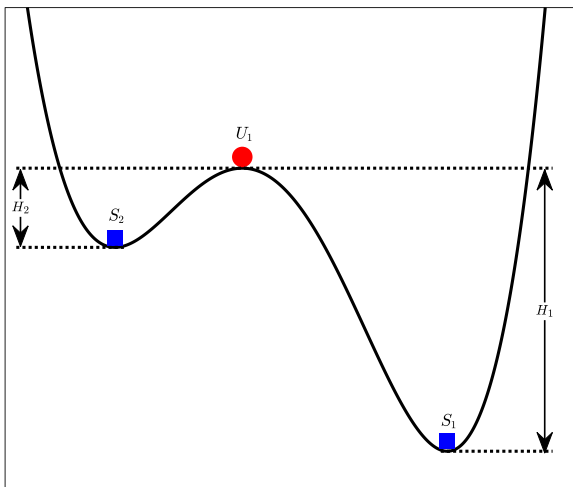


FIG. 5. Potential barrier heights H_1 and H_2 for the stable equilibrium points S_1 and S_2 , respectively. U_1 is the unstable equilibrium point.

the considered system are different, and we present some of their qualitative differences. The stable attractors at node 39 have small barrier heights when $d = 0.1$ [see Fig. 4(d)]. Therefore, the system can switch between the basin of attractions for noise intensity $d = 0.1$, which was not seen for the noise intensity $d = 0.001$. This shows that the system is in a nonequilibrium state at node 39. This is comparable to the dynamics of the stochastic system (2), where increasing noise intensity enhances the freedom of moving to other states [see Fig. 1].

The nonzero barrier height still exists for the noise level $d = 0.1$. For instance, we see a barrier between two valleys at node 4 in Fig. 4(e). Therefore, the noise intensity $d = 0.1$ does not put node 4 into a nonequilibrium state. In this case, nodes with connections can interact with other nodes, giving them the capacity to absorb external factors. This shows that it needs more potential (or noise intensity) to get into the nonequilibrium state, and hence, the stochastic system is more robust in the presence of low noise intensity. The same scenario happens for the other nodes having degrees more than the degree of node 4, e.g., see Fig. 4(f) for node 36. Therefore, the network model suggests that the same amount of drug applied to the brain causes different $A\beta$ and Ca^{2+} dynamics in different brain areas, which can be used for potential target-oriented therapeutic strategies aimed at restoring equilibrium and alleviating the pathological consequences associated with AD and related disorders.

It has been discussed that the barrier height does not remain the same for all the nodes in the brain connectome, even for low noise intensity levels. In addition, it has shown that the barrier height at the nodes varies with the noise intensity parameter d . However, the degree of a node plays a crucial role in the barrier height. Figure 6 depicts the barrier heights

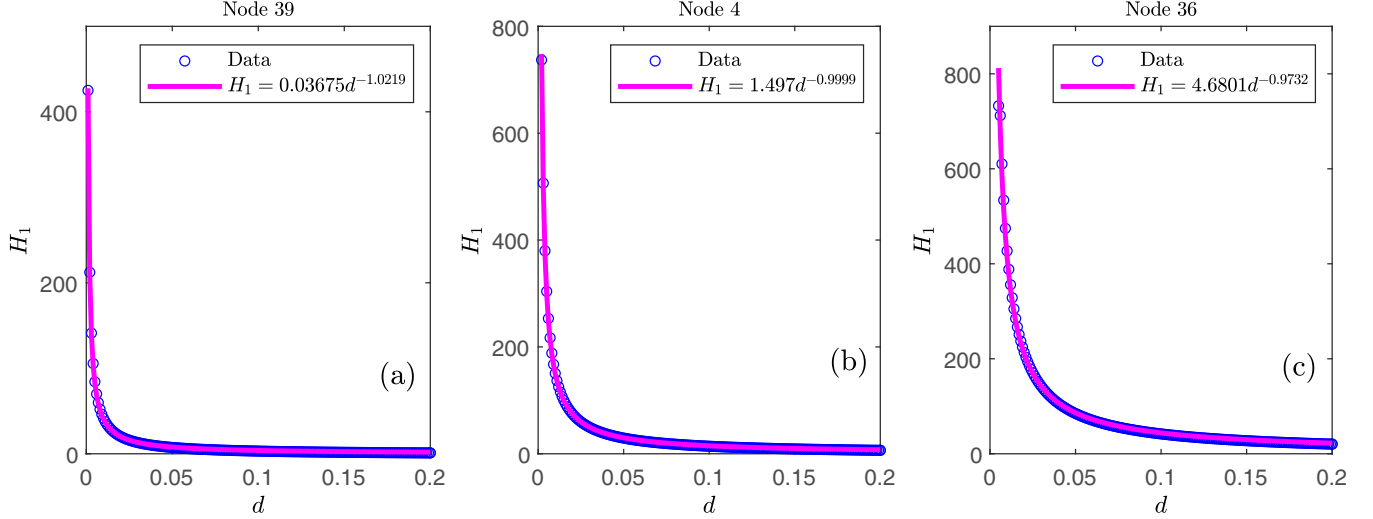


FIG. 6. The relationship between the noise intensity (d) and barrier height (H_1) for potential landscapes of the stationary states at different nodes in the brain connectome.

for nodes 39, 4, and 36 in the integrated brain connectome data for different noise intensity levels. These nodes are chosen due to their different degrees, which are increasing in order. This result shows that the barrier height of each basin of attractions decreases with increasing the noise intensity level. Therefore, less energy potential is required to overcome a transition state in the presence of high-intensity noise. On the other hand, the barrier height increases with the degree of the node, which shows the robustness depends on the degree of the node. In energy landscapes, nodes with high connectivity represent states with higher energy barriers to transitions due to increased constraints. As a result, more noise intensity is required for high-degree nodes if to change their state. Furthermore, the scaling factors of the power law fit show a stronger relationship between the noise intensity and the barrier height and, hence, with degrees. Nevertheless,

the system shows a remarkable difference in the barrier height if the degree of each node is not the same [see Fig. 7]. We have verified all the results with different brain connectome data, and no qualitative difference has been observed.

IV. CONCLUSIONS

This paper explored the stochastic interference of the interactions between $A\beta$ and Ca^{2+} in the brain. Nonlinearity typically implies multistability, many types of coexisting attractors, and complex shapes of the basins of attraction. Here, the deterministic model also exhibits multistability, and it has a saddle point and two stable attractors: healthy and pathological states. Each of these stable attractors has a basin of attraction, which is separated by the separatrix generated from the saddle point. One could apply the non-normal effect to see the short-term dynamics of the system when it gives long transients [40].

In general, the trajectories for the deterministic system do not cross separatrices; however, they can cross such borders in the presence of random disturbance, which gives a transition from one stable attractor to another. For the low noise intensity, the trajectories of the noise-induced system leave a deterministic attractor and lie in close proximity to a deterministic attractor. Confidence domains are the conventional method for achieving the geometric representation of such regions working with statistical estimate accuracy. The stochastic sensitivity analysis approach is applied to the stochastic system to find such proximity regions called confidential ellipses. However, a higher noise intensity can force the randomly distributed trajectories to leave these confidential ellipses and may continuously switch between the deterministic basin of attractions. This shows that a higher dose of a drug can be applied for AD patients to switch the pathological state to the healthy state, but some control is required to maintain such a healthy state for the rest of their lives.

We have developed a network model based on two ordinary differential equations to represent the interactions between intracellular $A\beta$ and Ca^{2+} in the brain connectome. The pro-

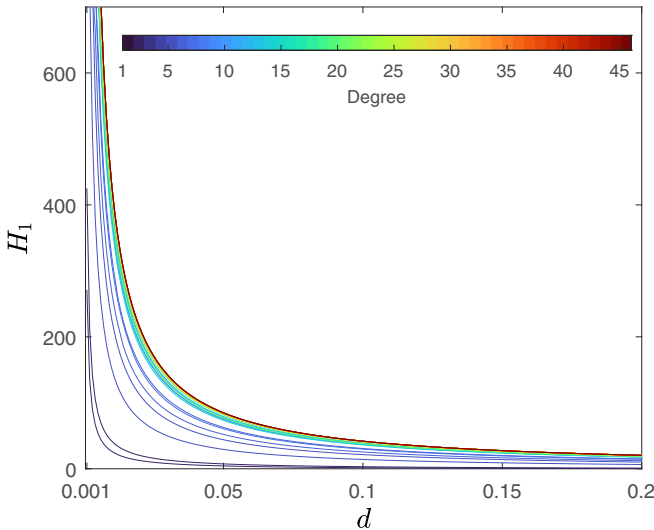


FIG. 7. The impact of noise intensity (d) on barrier height (H_1) for potential landscapes of the stationary states for various degree nodes. The degree of each node in the brain connectome is shown by the colorbar.

posed network model quantifies how individual nodes respond in the presence of stochastic noises based on biological features. The topic is essential because it concerns the linkage with the modern qualitative theory of deterministic dynamical systems with stochastic analysis. The probabilistic evolution approach is applied to characterize the dynamics locally at each node. We have shown that the potential landscape does not remain the same for each node in the brain connectome, and this is true for any amount of noise. Nevertheless, the degree of the nodes in the brain connectome plays an important role in the resultant potential landscape.

For a low noise intensity, the area of the valleys in the potential landscape is proportional to the degree of the node. These valleys are disjoint for the nodes with higher degrees. However, they do not remain isolated at some nodes in the presence of high noise intensity, and hence, the stochastic system shows nonequilibrium dynamics at those nodes. This indicates that the transformation of the pathological state to the healthy state can be done for such nodes in the brain with a high-potential drug. In addition, this high-potential drug can harm other brain regions as it may transfer to other brain regions through neural fiber tracts.

For a given noise intensity, the barrier height of a node gives the information on whether the node shows equilibrium or nonequilibrium dynamics. In addition, for a given stable

attractor, a larger barrier height means it is harder to escape from its basin of attraction, making it more resilient. For any node, the numerical simulation results show that the barrier height decreases with increasing noise intensity, which fits the power law with a negative exponent. The scaling factor and the exponent of the power law are not the same for all the nodes, and they depend on the degree of the node. Furthermore, we have observed that the barrier height is proportional to the degree of the nodes. This shows that the regions that are more connected require more potential drugs to switch to their current state, which can further change the state of the regions that are less connected. Therefore, starting from a lower connected region in a region-oriented therapy is a better option for curing the disease.

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- [1] H. Hippius and G. Neundörfer, The discovery of Alzheimer's disease, *Dialogues Clin. Neurosci.* **5**, 101 (2003).
- [2] M. F. Mendez, Early-onset Alzheimer disease and its variants, *Continuum: Lifelong Learn. Neurol.* **25**, 34 (2019).
- [3] S. Duong, T. Patel, and F. Chang, Dementia: What pharmacists need to know, *Can. Pharm. J.* **150**, 118 (2017).
- [4] K. G. Yiannopoulou and S. G. Papageorgiou, Current and future treatments in Alzheimer disease: an update, *J. Cent. Nerv. Syst. Dis.* **12**, 117957352090739 (2020).
- [5] D. Jeremic, L. Jiménez-Díaz, and J. D. Navarro-López, Past, present and future of therapeutic strategies against amyloid- β peptides in Alzheimer's disease: A systematic review, *Ageing Res. Rev.* **72**, 101496 (2021).
- [6] V. W. Chow, M. P. Mattson, P. C. Wong, and M. Gleichmann, An overview of APP processing enzymes and products, *NeuroMol. Med.* **12**, 1 (2010).
- [7] G.-f. Chen, T.-h. Xu, Y. Yan, Y.-r. Zhou, Y. Jiang, K. Melcher, and H. E. Xu, Amyloid beta: structure, biology and structure-based therapeutic development, *Acta Pharmacol. Sin.* **38**, 1205 (2017).
- [8] A. Serrano-Pozo, M. P. Frosch, E. Masliah, and B. T. Hyman, Neuropathological alterations in Alzheimer disease, *Cold Spring Harbor Perspect. Med.* **1**, a006189 (2011).
- [9] Y. Zhang, H. Chen, R. Li, K. Sterling, and W. Song, Amyloid β -based therapy for alzheimer's disease: Challenges, successes and future, *Signal Transduct. Target. Ther.* **8**, 248 (2023).
- [10] R. Tenchov, J. M. Sasso, and Q. A. Zhou, Alzheimer's disease: Exploring the landscape of cognitive decline, *ACS Chem. Neurosci.* **15**, 3800 (2024).
- [11] M. J. Berridge, Neuronal calcium signaling, *Neuron* **21**, 13 (1998).
- [12] E. M. Kawamoto, C. Vivar, and S. Camandola, Physiology and pathology of calcium signaling in the brain, *Front. Pharmacol.* **3**, 22359 (2012).
- [13] G. Zündorf and G. Reiser, Calcium dysregulation and homeostasis of neural calcium in the molecular mechanisms of neurodegenerative diseases provide multiple targets for neuroprotection, *Antioxid. Redox Signaling* **14**, 1275 (2011).
- [14] C. Supnet and I. Bezprozvanny, Neuronal calcium signaling, mitochondrial dysfunction, and alzheimer's disease, *J. Alzheimers Dis.* **20**, S487 (2010).
- [15] M. Calvo-Rodriguez, E. K. Kharitonova, and B. J. Bacskaï, Therapeutic strategies to target calcium dysregulation in Alzheimer's disease, *Cells* **9**, 2513 (2020).
- [16] D. Baracaldo-Santamaría, S. S. Avendaño-Lopez, D. F. Ariza-Salamanca, M. Rodríguez-Giraldo, C. A. Calderon-Ospina, R. E. González-Reyes, and M. O. Nava-Mesa, Role of calcium modulation in the pathophysiology and treatment of Alzheimer's disease, *Int. J. Mol. Sci.* **24**, 9067 (2023).
- [17] M. Cameron, O. Kékesi, J. W. Morley, J. Tapson, P. P. Breen, A. Van Schaik, and Y. Buskila, Calcium imaging of AM dyes following prolonged incubation in acute neuronal tissue, *PLoS ONE* **11**, e0155468 (2016).
- [18] L. Kosten, S. A. Emmi, S. Missault, and G. A. Keliris, Combining magnetic resonance imaging with readout and/or perturbation of neural activity in animal models: Advantages and pitfalls, *Front. Neurosci.* **16**, 938665 (2022).
- [19] M. P. Mattson, B. Cheng, D. Davis, K. Bryant, I. Lieberburg, and R. E. Rydel, beta-Amyloid peptides destabilize calcium homeostasis and render human cortical neurons vulnerable to excitotoxicity, *J. Neurosci.* **12**, 376 (1992).

- [20] K. N. Green and F. M. LaFerla, Linking calcium to $A\beta$ and Alzheimer's disease, *Neuron* **59**, 190 (2008).
- [21] J. De Caluwé and G. Dupont, The progression towards Alzheimer's disease described as a bistable switch arising from the positive loop between amyloids and Ca^{2+} , *J. Theor. Biol.* **331**, 12 (2013).
- [22] P.-P. Guan, L.-L. Cao, and P. Wang, Elevating the levels of calcium ions exacerbate alzheimer's disease via inducing the production and aggregation of β -amyloid protein and phosphorylated tau, *Int. J. Mol. Sci.* **22**, 5900 (2021).
- [23] J. Gao, J. Ji, Y. Liu, F. Zhang, R. Wang, and H. Wang, A confidence ellipse analysis for stochastic dynamics model of Alzheimer's disease, *Nonlinear Dyn.* **111**, 13513 (2023).
- [24] L. S. Tsimring, Noise in biology, *Rep. Prog. Phys.* **77**, 026601 (2014).
- [25] R. Hancock, K. R. Pugh, and F. Hoeft, Neural noise hypothesis of developmental dyslexia, *Trends Cognit. Sci.* **21**, 434 (2017).
- [26] Y. Zheng, S. Tang, H. Zheng, X. Wang, L. Liu, Y. Yang, Y. Zhen, and Z. Zheng, Noise improves the association between effects of local stimulation and structural degree of brain networks, *PLoS Comput. Biol.* **19**, e1010866 (2023).
- [27] N. G. Van Kampen, *Stochastic Processes in Physics and Chemistry* (Elsevier, Amsterdam, The Netherlands, 1992), Vol. 1.
- [28] M. I. Freidlin and A. D. Wentzell, *Random Perturbations of Dynamical Systems* (Springer New York, NY, 1998), pp. 15–43.
- [29] C. W. Gardiner *et al.*, *Handbook of stochastic methods* (Springer-Verlag Berlin, Germany, 1985), Vol. 3.
- [30] I. Bashkirtseva, L. Ryashko, and T. Ryazanova, Method of confidence domains in the analysis of noise-induced extinction for tritrophic population system, *Eur. Phys. J. B* **90**, 1 (2017).
- [31] M. I. Freidlin and A. D. Wentzell, *Random Perturbations of Dynamical Systems* (Springer-Verlag Berlin, Germany, 2012).
- [32] T. B. Thompson, P. Chaggar, E. Kuhl, A. Goriely, and A. D. N. Initiative, Protein-protein interactions in neurodegenerative diseases: A conspiracy theory, *PLoS Comput. Biol.* **16**, e1008267 (2020).
- [33] S. Pal and R. Melnik, Nonlocal models in the analysis of brain neurodegenerative protein dynamics with application to Alzheimer's disease, *Sci. Rep.* **12**, 7328 (2022).
- [34] L. Ye, J. Feng, and C. Li, Controlling brain dynamics: Landscape and transition path for working memory, *PLoS Comput. Biol.* **19**, e1011446 (2023).
- [35] J. Wang, L. Xu, and E. Wang, Potential landscape and flux framework of nonequilibrium networks: Robustness, dissipation, and coherence of biochemical oscillations, *Proc. Natl. Acad. Sci. USA* **105**, 12271 (2008).
- [36] I. Dornic, H. Chaté, and M. A. Munoz, Integration of langevin equations with multiplicative noise and the viability of field theories for absorbing phase transitions, *Phys. Rev. Lett.* **94**, 100601 (2005).
- [37] S. di Santo, P. Villegas, R. Burioni, and M. A. Muñoz, Landau–Ginzburg theory of cortex dynamics: Scale-free avalanches emerge at the edge of synchronization, *Proc. Natl. Acad. Sci. USA* **115**, E1356 (2018).
- [38] <https://braingraph.org>.
- [39] B. Szalkai, C. Kerepesi, B. Varga, and V. Grolmusz, High-resolution directed human connectomes and the consensus connectome dynamics, *PLoS ONE* **14**, e0215473 (2019).
- [40] S. di Santo, P. Villegas, R. Burioni, and M. A. Muñoz, Non-normality, reactivity, and intrinsic stochasticity in neural dynamics: a non-equilibrium potential approach, *J. Stat. Mech.* (2018) 073402.
- [41] www.sharcnet.ca.
- [42] www.alliancecan.ca.